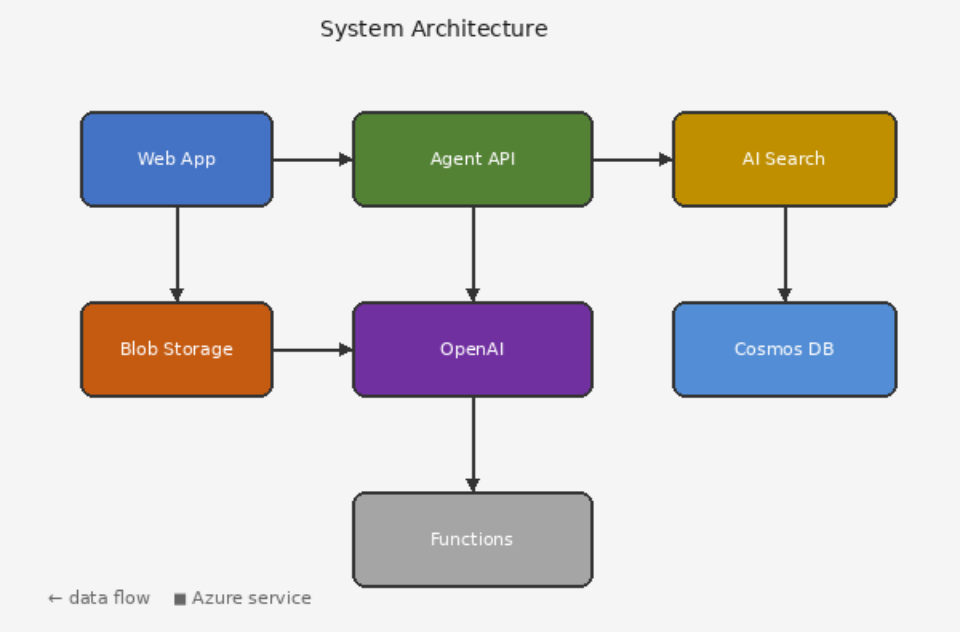
Azure Knowledge Base Architecture Guide

Introduction

This document describes the architecture of the Azure Knowledge Base system. The system uses Azure AI Search for retrieval-augmented generation (RAG) and Azure OpenAI Service for natural language understanding. For more details, see the MarkItDown GitHub repository.

System Architecture

The following diagram shows the high-level architecture of the system:



Component Overview

Component	Service	Purpose	SKU
Search Index	Azure AI Search	Vector + keyword search	Standard S1
LLM	Azure OpenAI	GPT-4.1 for generation	Standard
Storage	Azure Blob Storage	KB article files	Standard LRS
Identity	Azure Entra ID	Authentication & RBAC	P1
Monitoring	Application Insights	Telemetry & logging	Pay-as-you-go

Deployment Options

Container Apps Hosting

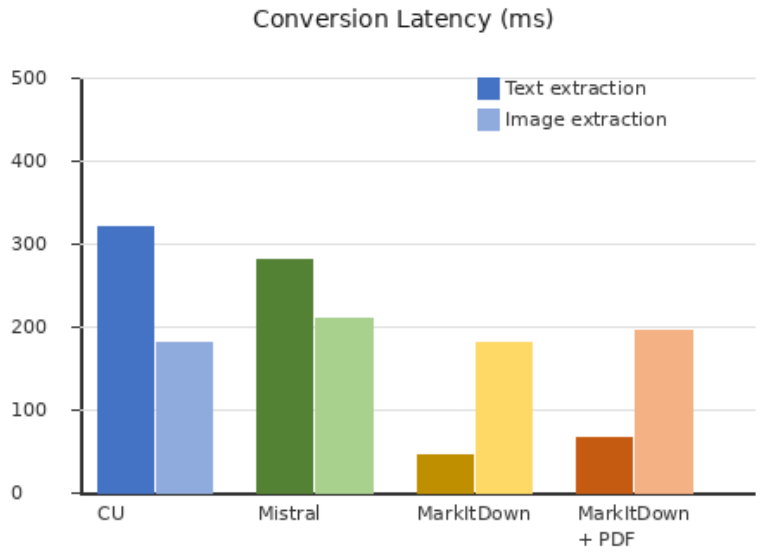
The agent is deployed as an Azure Container App with managed identity enabled. Environment variables are configured via Bicep templates, and secrets are stored in Azure Key Vault.

Scaling Configuration

Auto-scaling is configured based on HTTP request concurrency. The minimum replica count is set to 1 for development environments and 3 for production.

Performance Metrics

The chart below compares conversion latency across backends:

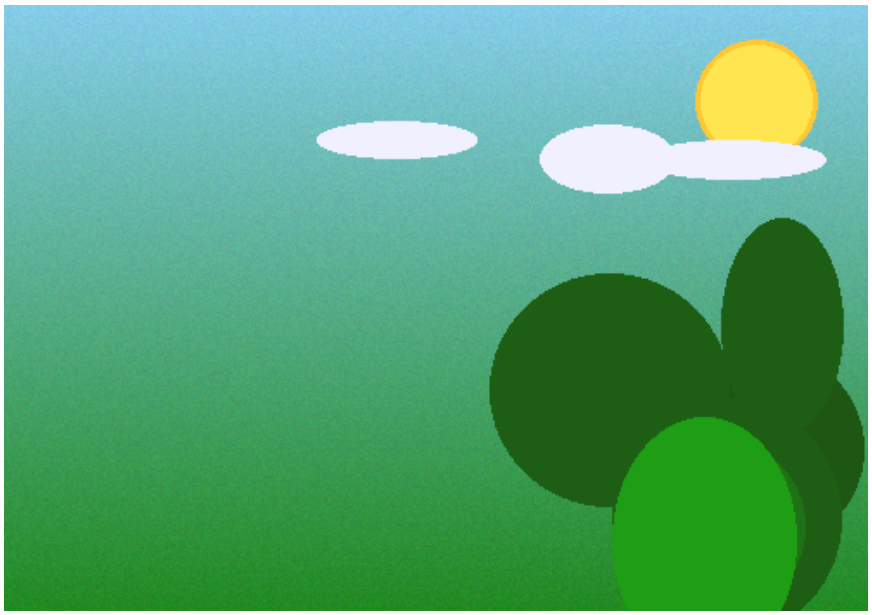


Key Features

• Retrieval-augmented generation with hybrid search (vector + keyword)
• Image-grounded responses using GPT-4.1 vision capabilities
• Per-request context isolation using Python ContextVar
• Managed identity for all service-to-service authentication
• Infrastructure as Code using Bicep templates

Sample Photo

Photo-like images with complex color gradients and organic shapes test whether image extraction preserves color fidelity and detail:



API Endpoints

Endpoint	Method	Description
/api/chat	POST	Send a chat message to the agent
/api/health	GET	Health check endpoint
/api/convert-markitdown	POST	Convert HTML to Markdown
/api/index	POST	Index articles into AI Search

Azure Resource Inventory

The following table lists all Azure resources deployed for the KB Agent system. This table is intentionally long (30 rows) to test cross-page table handling.

#	Resource Name	Type	Region	Status	Monthly Cost
1	kb-agent-container-app-01	Container App	East US 2	Running	$45.00
2	kb-agent-cosmos-db-02	Cosmos DB	West US 3	Provisioned	$120.50
3	kb-agent-ai-search-03	AI Search	Central US	Scaling	$89.99
4	kb-agent-openai-04	OpenAI	North Europe	Updating	$210.00
5	kb-agent-blob-storage-05	Blob Storage	West Europe	Healthy	$15.75
6	kb-agent-key-vault-06	Key Vault	East US 2	Running	$32.40
7	kb-agent-app-insights-07	App Insights	West US 3	Provisioned	$8.50
8	kb-agent-log-analytics-08	Log Analytics	Central US	Scaling	$55.20
9	kb-agent-container-registry-09	Container Registry	North Europe	Updating	$178.00

#	Resource Name	Type	Region	Status	Monthly Cost
10	kb-agent-virtual-network-10	Virtual Network	West Europe	Healthy	$0.00
11	kb-agent-container-app-11	Container App	East US 2	Running	$45.00
12	kb-agent-cosmos-db-12	Cosmos DB	West US 3	Provisioned	$120.50
13	kb-agent-ai-search-13	AI Search	Central US	Scaling	$89.99
14	kb-agent-openai-14	OpenAI	North Europe	Updating	$210.00
15	kb-agent-blob-storage-15	Blob Storage	West Europe	Healthy	$15.75
16	kb-agent-key-vault-16	Key Vault	East US 2	Running	$32.40
17	kb-agent-app-insights-17	App Insights	West US 3	Provisioned	$8.50
18	kb-agent-log-analytics-18	Log Analytics	Central US	Scaling	$55.20
19	kb-agent-container-registry-19	Container Registry	North Europe	Updating	$178.00
20	kb-agent-virtual-network-20	Virtual Network	West Europe	Healthy	$0.00
21	kb-agent-container-app-21	Container App	East US 2	Running	$45.00
22	kb-agent-cosmos-db-22	Cosmos DB	West US 3	Provisioned	$120.50
23	kb-agent-ai-search-23	AI Search	Central US	Scaling	$89.99
24	kb-agent-openai-24	OpenAI	North Europe	Updating	$210.00
25	kb-agent-blob-storage-25	Blob Storage	West Europe	Healthy	$15.75
26	kb-agent-key-vault-26	Key Vault	East US 2	Running	$32.40
27	kb-agent-app-insights-27	App Insights	West US 3	Provisioned	$8.50
28	kb-agent-log-analytics-28	Log Analytics	Central US	Scaling	$55.20
29	kb-agent-container-registry-29	Container Registry	North Europe	Updating	$178.00
30	kb-agent-virtual-network-30	Virtual Network	West Europe	Healthy	$0.00

References

- Azure AI Search documentation
- Azure OpenAI Service
- MarkItDown on GitHub
- Azure Container Apps
- Bicep documentation